Paired cuts for a selected set of colours

Human Compression project: L044 continuation

Let $H = \langle s_0, \ldots, s_{r-1} \rangle$ act transitively on a finite set U of size m. Each s_i is an involution or the identity, and $[s_i, s_j] = 1$ for $|i - j| > 1$. For every index j, put $H_j = \langle s_i : i \neq j \rangle$. An edge of colour j is eligible when its endpoints lie in different orbits of this *full local* omission subgroup H_j. Fix any colour set $F \subseteq \{0, \ldots, r-1\}$. It need not be an interval, and $H_F = \langle s_j : j \in F \rangle$ need not be transitive. Suppose every colour in F has at least two eligible edges, and select exactly two of each on the full vertex set U.

Theorem 1. *The bound $2|F| \leq m$ holds. If $2|F| = m$, there is a connected paired selection using just the colours in F, and H_F is transitive on U. Consequently, if H_F is intransitive, then*

$$2|F| \leq m - 1. \tag{1}$$

No intersection property, parity or primitivity is required.

Proof. We retain the full accepted proofs of *Paired orbit cuts in string actions* and *Localisation in the paired-cut equality case*. We check every place where selecting fewer than all colours could change their argument.

Orbit-cut exchange, the selected-square block lemma and long-cycle reduction use only the full local H_j cuts and the fact that each selected colour has exactly two edges. They therefore give a square-only selection. Ordinary slides between squares use colours of selected connector edges, so all colours they exchange belong to F. If such a connector begins with the middle colour i of a square of colours $i-1, i+1$, then i is itself selected. The global middle-colour move retains this square and transports every other selected edge by s_i. Every transported colour other than i commutes with s_i, except the two square colours, whose selected edges have all been retained; i-edges map to themselves. Eligibility is preserved by $s_i \in H_j$ for $j \neq i$. Thus the complete minimum-square and minimum-distance proof still gives at most one square per component.

If c, q are the numbers of components and squares in such a selection, including isolated vertices among the components, its exact defect is $2|F| = m - c + q$. Hence $2|F| \leq m$. In equality, every minimum-square selection has exactly one square in each component. We now adapt the complete equality connectedness proof.

First, every eligible edge whose colour belongs to a selected square lies in that square's component. An edge in another component would have a connecting selected path avoiding its colour. An edge joining components could replace a square edge, destroying that square and joining components without a new cycle. These are the two contradictions in the accepted cycle-colour localisation lemma.

Next consider a full graph edge of any colour k, selected or not, incident with a square vertex. If s_k commutes with a square colour j, transporting its eligible j-edge by $s_k \in H_j$ and applying localisation keeps the new endpoint in that component. The only other case has square colours $i-1, i+1$ and $k = i$. Here i may be *unselected*. Nevertheless the same global exchange is valid: retain the square and transport all other selected edges by s_i. If $i \notin F$ there are simply no selected i-edges to account for. All other selected colours outside the square commute with s_i, and their eligibility is preserved.

Viewed on the transported graph, the exchange removes the transported square and inserts the original square. The removed square splits its old unicyclic component into four trees. Every

inserted edge lies in the inserted full square, so creates no other simple cycle. The number of squares is unchanged. Each inserted vertex must lie in those four trees: if it lay in another unchanged component, the new component would have two squares, contrary to the minimum-square property. All four trees must be rejoined, or a tree component would remain in equality. Thus the new component partition is exactly the transported old partition, and the incident full edge stays in the original square's component. This is precisely the earlier full-neighbour and partition argument; the missing middle colour causes no exception.

If the selection were disconnected, full H transitivity would give a cross-edge between components. Over all minimum-square selections, choose a shortest connector consisting of selected tails from two squares and exactly one full cross-edge. Both tails are nonempty by the full-neighbour property. The accepted ordinary-slide argument shortens either tail unless its first colour is the middle colour of its square. Its repeated-colour exception is ruled out by cycle-colour localisation, and ordinary slides preserve the component vertex sets, exactly as before.

Both first colours must therefore be middle colours. The two squares have disjoint colour sets. The cross-edge colour x is outside the colour set of at least one of them; choose that endpoint, with middle colour i. Then s_i commutes with s_x, including $x = i$. The global middle exchange transports the cross-edge to an actual x-edge and transports every connector edge after the first. The transported component partition makes it a cross-edge still, and the new tail starts on the retained square and is shorter by one. The middle colour here belongs to F, since it was a selected tail edge. This gives the same final contradiction as in the accepted connectedness proof. Thus the equality selection is connected. Its edges all have colours in F, so H_F is transitive. The integer strict bound (1) follows. □

Corollary 2 (Exact local component interface)**.** *Let O be an orbit of a tail in an ambient string action, and let H be that tail's full restricted action on O. Suppose that two edges of every colour in F cross orbits of the actual full local omission H_j. Then $|F| \leq |O|/2$. If some index $k \notin F$ has H_k intransitive, then $|F| \leq (|O| - 1)/2$.*

Proof. Apply Theorem 1, padding the other indices with identity restrictions when needed. In the second case $H_F \leq H_k$ is intransitive. No identification with an ambient omission is used. □

Attribution and limits. This supplies the selected-colour local rank interface appearing in Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, Propositions 5.27 and 5.31, physical pages 35–36 and 42–43. It does not prove that any particular source component satisfies its local eligibility or intransitivity hypotheses. Ambient omission eligibility and full local omission eligibility are distinct: neither is silently substituted for the other. It also asserts no sum of orbit deficits stronger than (1). The root-supplied global middle move and its complete distinct review remain bound through the paid predecessor manuscripts. The present change is the explicit selected-colour scope argument. There is no new census, formalization, source-wide recovery, original credit or question credit. Connected-square rigidity and the source's later component classification remain open obligations.